# Ben Nguyen

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## Education

**University of Florida** Gainesville, FL
*B.S. Computer Engineering GPA: 3.65* *Expected Dec 2027*
*Coursework:* Data Structures & Algorithms, Microprocessor Applications, Digital Logic, Circuits I, AI for Virtual Environments

## Technical Skills

**Languages:** C, C++, C#, Java, Python, JavaScript, AVR Assembly, VHDL, SQL
**Embedded & Hardware:** ESP32, Raspberry Pi Pico, Arduino, Altium Designer, I2S, Quartus Prime, SolidWorks
**Software & Frameworks:** Spring Boot, React, gRPC, REST APIs, WebSockets, PostgreSQL, Godot, Git
**Certificates:** Siemens Expedite Microcredential, AutoCAD Inventor

## Projects

**Portable Bluetooth Speaker** – *ESP32, C++, Altium Designer, I2S, SolidWorks, PCB Design*
- Designed a custom 2-layer PCB in Altium integrating an ESP32 and dual MAX98357A class-D amplifiers, battery-powered.
- Developed C++ firmware (Arduino framework) streaming Bluetooth audio over the I2S digital protocol to dual 5W speakers, with hardware interrupts and debouncing for tactile play/pause push-buttons.
- Programmed an LCD to render real-time system status and playback metadata; modeled and 3D-printed an enclosure to house the PCB, speakers, display, and controls using SolidWorks.

**Full-Stack E-Commerce Marketplace** – *Java, Spring Boot, C++, gRPC, React, PostgreSQL*
- Designed and developed an e-commerce marketplace using React, Spring Boot, and PostgreSQL, featuring a custom C++ remote procedure call (gRPC) microservice for real-time dynamic pricing calculations.
- Eliminated a 96% order failure rate during high-traffic load testing by implementing optimistic locking.
- Delivered real-time order notifications via STOMP-over-WebSocket, validated the system with Mockito unit tests.

**Robocode Tournament AI** – *Java, Robocode API, Finite State Machines, Minimum-Risk Movement*
- Placed 1st among 26 competitors with an autonomous Java tank AI, outscoring the benchmark bot by 91%; a finite-state machine (scan/attack/evade) governed target selection, firepower, and movement policy.
- Built a minimum-risk movement engine scoring 300 sampled destinations per re-position against an inverse-square risk field weighted by relative enemy energy and firing angle, minimizing exposure to incoming fire.
- Dodged shots by detecting enemy fire from per-tick energy drops; led targets with linear-intercept aiming.

**FishSchooling Simulation** – *C#, Godot Engine, Git*
- Implemented flocking (alignment, cohesion, separation) on an interface-based steering system to model autonomous 2D agents; collaborated with 3 developers via Git branching and pull requests.
- Built a dynamic UI exposing fish-type-specific parameter sliders, enabling real-time behavior tuning across 5 creature types in a predator–prey ecosystem.

## Experience

**Machine Intelligence Laboratory (MIL), University of Florida** Gainesville, FL
*Electrical Team Member* *August 2026 – Present*
- Developed the light board for the TerraGator rover, programming a Raspberry Pi Pico to drive AL8843SP-13 LED drivers via PWM for indicators and headlights.
- Researched PCB design practices, including component selection and layout, to inform team board designs; report progress and blockers in team standups.

**UF Association of Computer Engineers** Gainesville, FL
*Workshop Lead* *March 2026 – Present*
- Led brainstorming of events that provided opportunities to learn about computer engineering and its applications.
- Improved club engagement by hosting research lab tours and technical workshops; coordinated with presenters and staff to ensure seamless event execution.

**Kyoto University — Cross-Cultural Engineering Design Program** Kyoto, Japan
*Exchange Student & Networking Event Coordinator* *September 2025 – December 2025*
- Built FishSchooling with a mixed Japanese–international team, communicating across language barriers.
- Organized networking lunches averaging 60 attendees, raising Japanese student attendance by 40%.